

# RK vs NN proton reconstruction on breakup g4output

## Head-to-head evaluation

2026-06-18

### Abstract

This report evaluates proton reconstruction quality on the d+Sn breakup sample (ypol + zpol, 176 cells) using two backends: RK (Runge-Kutta least-squares fitter, three-point-free mode, run on spana03) and NN (6-128-128-64 MLP, formal\_B115T3deg/20260420\_184345 model, run locally). Both reconstructions derive from the same Geant4 g4output, enabling per-event entry-index alignment. Three error sets are reported for protons (RK-vs-truth, NN-vs-truth, RK-vs-NN), plus truth-vs-reco residuals for neutrons (NEBULA  $\beta$ -method, identical for both backends). Sample: 3301419 truth-proton events total.

## 1 Data & method

**RK reco:** spana03 data/reconstruction/dbreak\_elastic/, 176 files, produced 2026-05-09 via `run_dbreak_elastic_pipeline.py stage_reco` with `--backend rk --rk-fit-mode three-point-free --max-iterations 40`.

**NN reco:** local data/reconstruction/breakup\_nn\_20260503/, 176 files, produced 2026-05-03 via `01_run_nn_reco.sh` with the clean production model `formal_B115T3deg/20260420_184345`.

**g4output source:** identical for both backends (verified by file size 285252607 bytes, timestamp 2026-05-01 20:16:46 +0900). Per-event entry-index alignment is valid 1:1.

**Frame convention:** target frame (beam-as-Z) throughout. Truth is already target-frame; both backends' reco is rotated to target frame at write time via `RotateRecoResultToTargetFrame` (`apps/run_reconstruction/main.cc:629-630`). No additional rotation in this analysis.

**Multi-proton events:** pick candidate whose  $|p|$  is closest to truth (matches `evaluate_target_momentum_reco.cc`).

## 2 Proton reconstruction — RK vs NN vs truth

### 2.1 Global residuals

| Pol  | Backend | $\Delta p_x$                                        | $\Delta p_y$                                     |
|------|---------|-----------------------------------------------------|--------------------------------------------------|
| ypol | RK      | MAE=11.4, RMSE=39.7, Bias=-6.8, P95=30.7, N=2150443 | MAE=1.6, RMSE=3.5, Bias=-0.1, P95=4.1, N=2150443 |
| ypol | NN      | MAE=6.1, RMSE=13.3, Bias=-1.8, P95=14.4, N=2150443  | MAE=1.3, RMSE=2.4, Bias=+0.5, P95=3.1, N=2150443 |
| zpol | RK      | MAE=12.0, RMSE=46.8, Bias=-7.4, P95=35.3, N=101662  | MAE=1.7, RMSE=4.8, Bias=-0.2, P95=3.1, N=101662  |
| zpol | NN      | MAE=7.1, RMSE=13.7, Bias=-2.6, P95=22.2, N=101662   | MAE=1.4, RMSE=3.0, Bias=+0.6, P95=2.1, N=101662  |

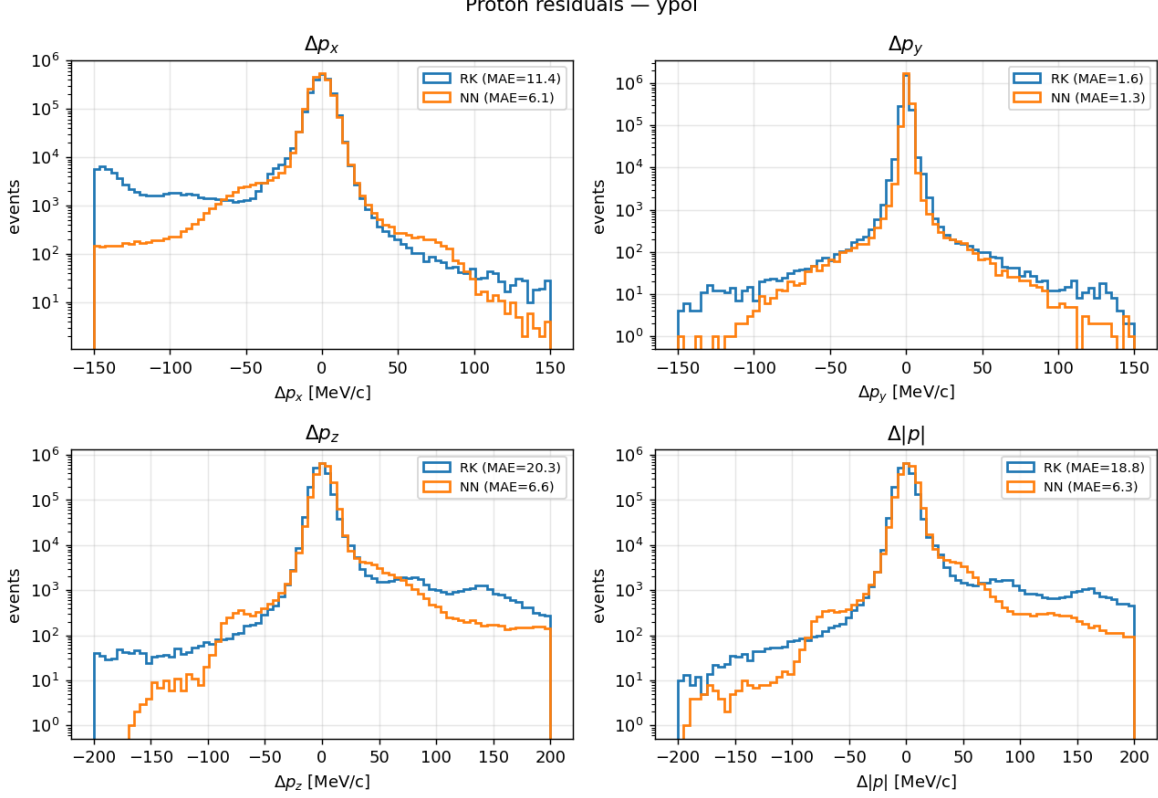


Figure 1: Proton residual distributions, ypol.

## 2.2 Per-event agreement (RK vs NN)

For each event, compare  $|p_{RK} - p_{truth}|$  vs  $|p_{NN} - p_{truth}|$ . Points below the  $y = x$  line are events where RK is closer to truth; above means NN is closer.

## 2.3 Per-cell breakdown

## 2.4 RK uncertainty calibration (pull)

Pull =  $(|p|_{RK} - |p|_{truth})/\sigma_p$ . Should be  $\sim \mathcal{N}(0, 1)$  if Fisher uncertainties are calibrated.

## 3 Neutron reconstruction — truth vs reco

Neutron momentum is reconstructed by the NEBULA  $\beta$ -method in both RK and NN backends (the backend choice only affects proton reconstruction). Residuals below are therefore single-backend (NN file used).

## 4 Conclusion

(To be filled in after inspecting the figures.)

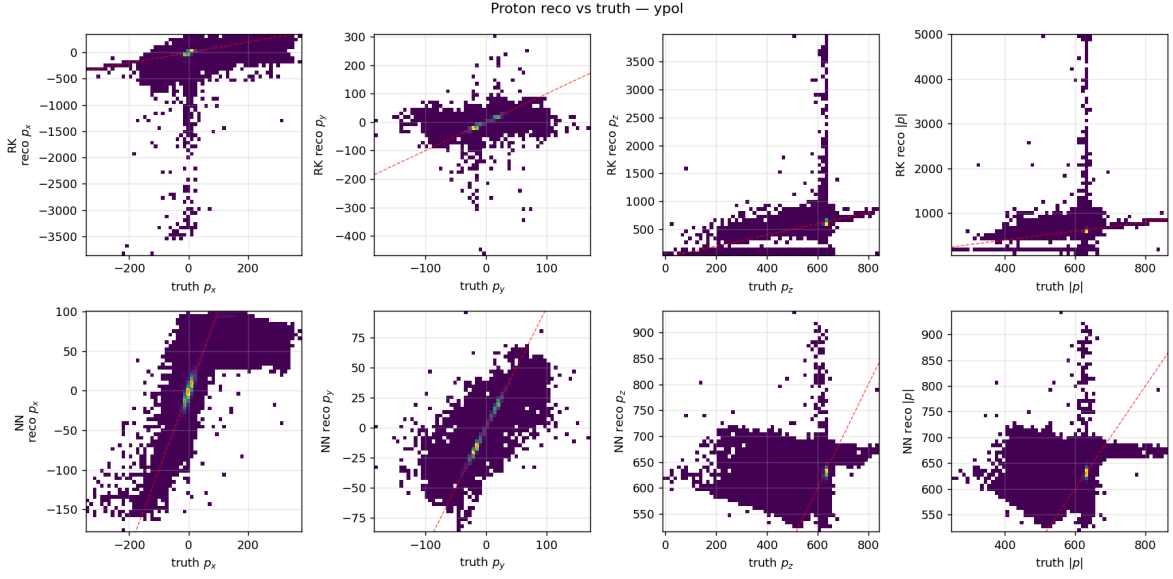


Figure 2: Proton reco vs truth 2D scatter, ypol.

## A Per-cell numeric table

(See `summary.json` for full per-cell numeric breakdown.)

Proton residuals — zpol

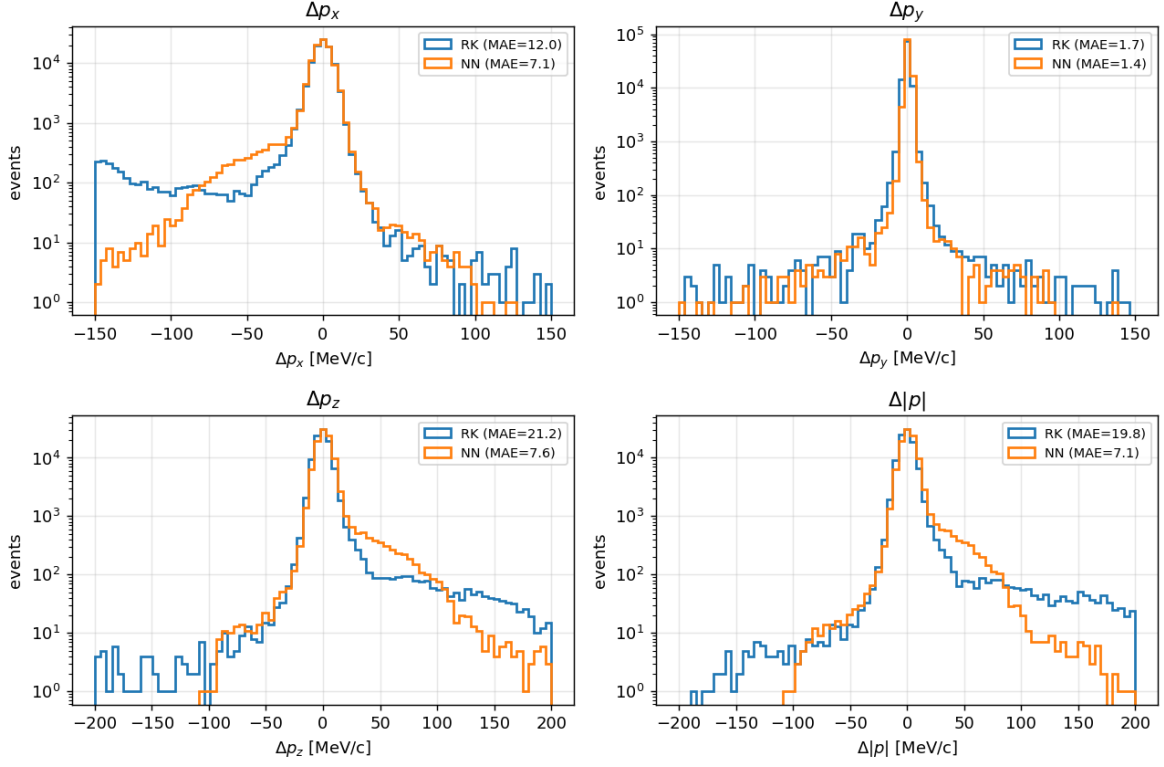


Figure 3: Proton residual distributions, zpol.

Proton reco vs truth — zpol

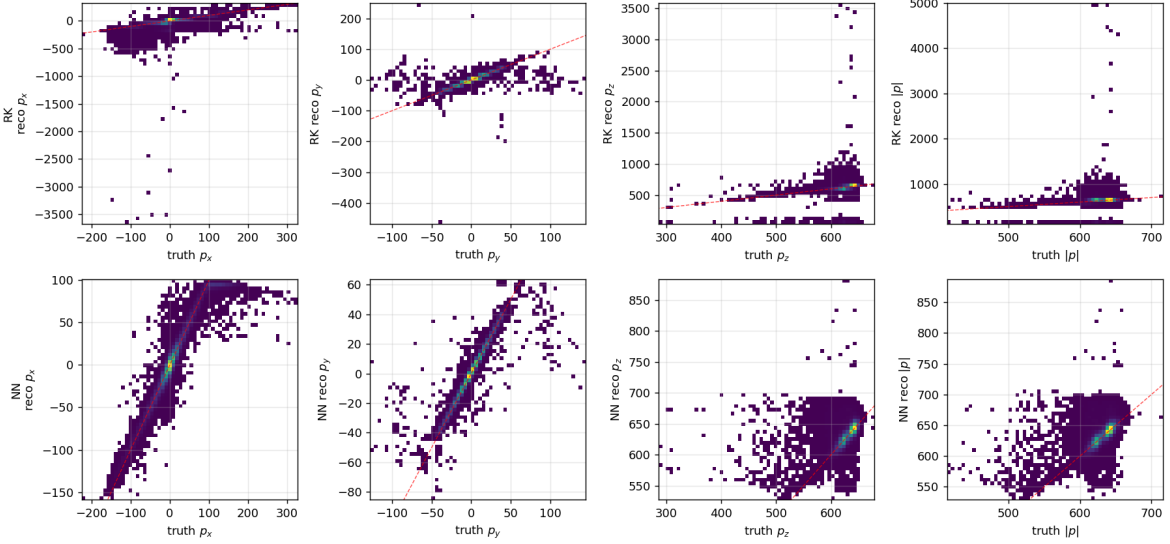


Figure 4: Proton reco vs truth 2D scatter, zpol.

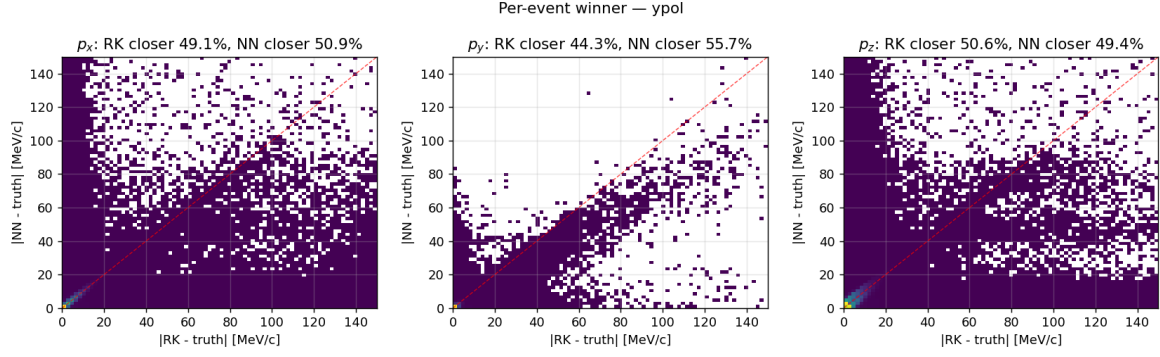


Figure 5: Per-event winner, ypol.

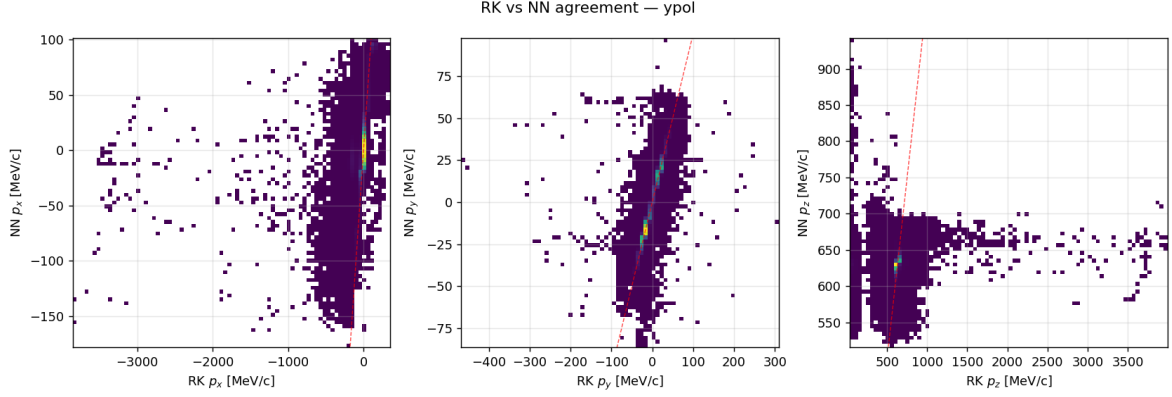


Figure 6: RK vs NN direct agreement, ypol.

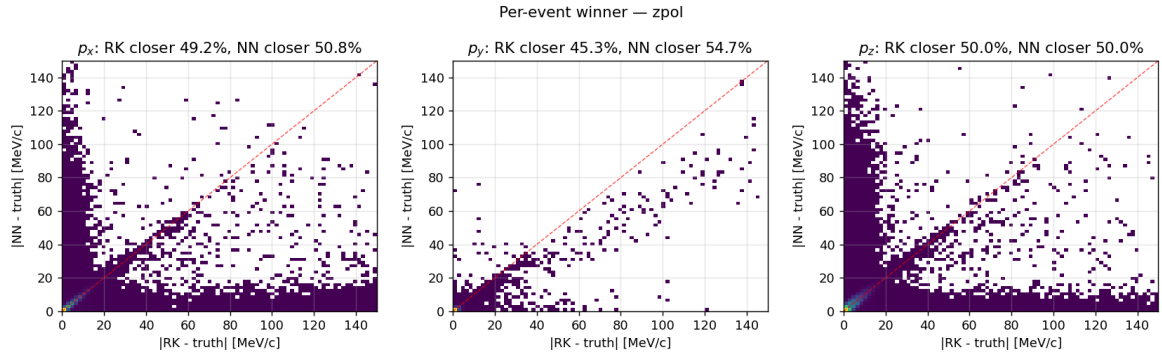


Figure 7: Per-event winner, zpol.

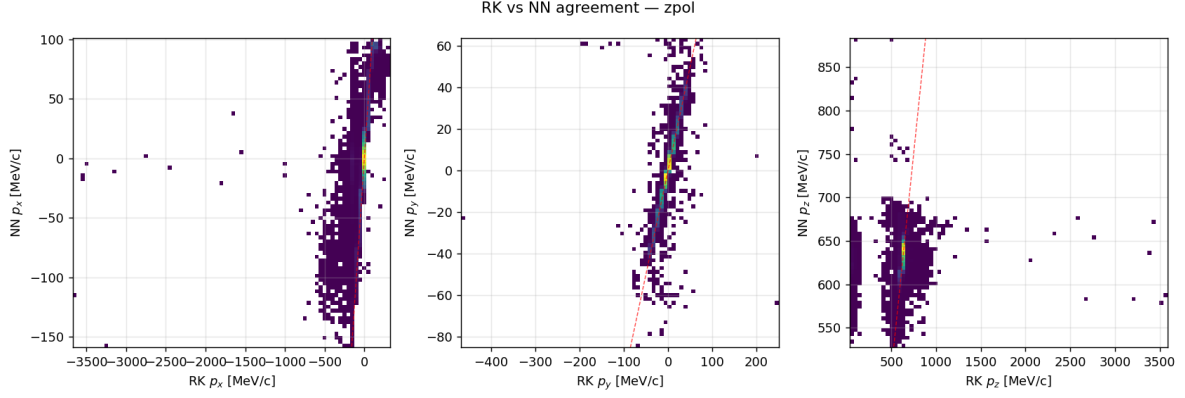


Figure 8: RK vs NN direct agreement, zpol.

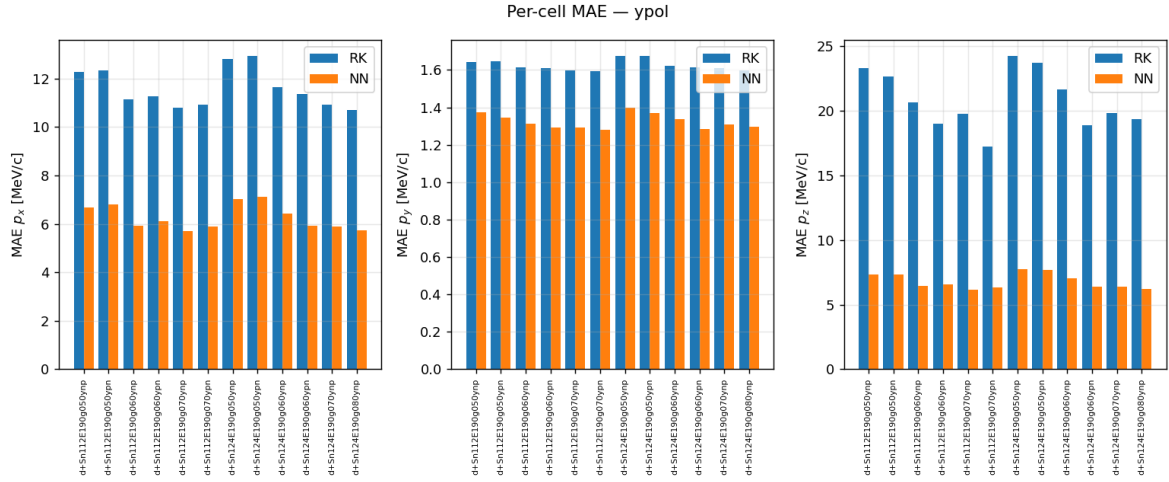


Figure 9: Per-cell MAE, ypol.

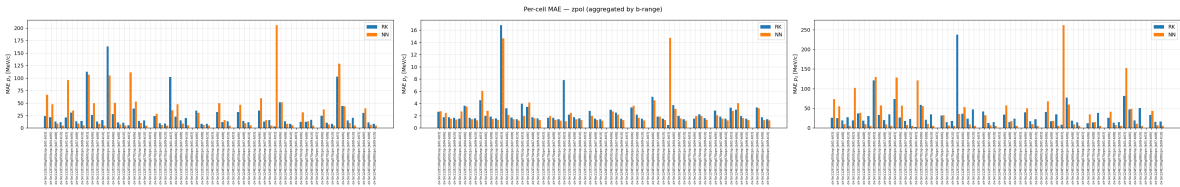


Figure 10: Per-cell MAE, zpol (aggregated by b-range).

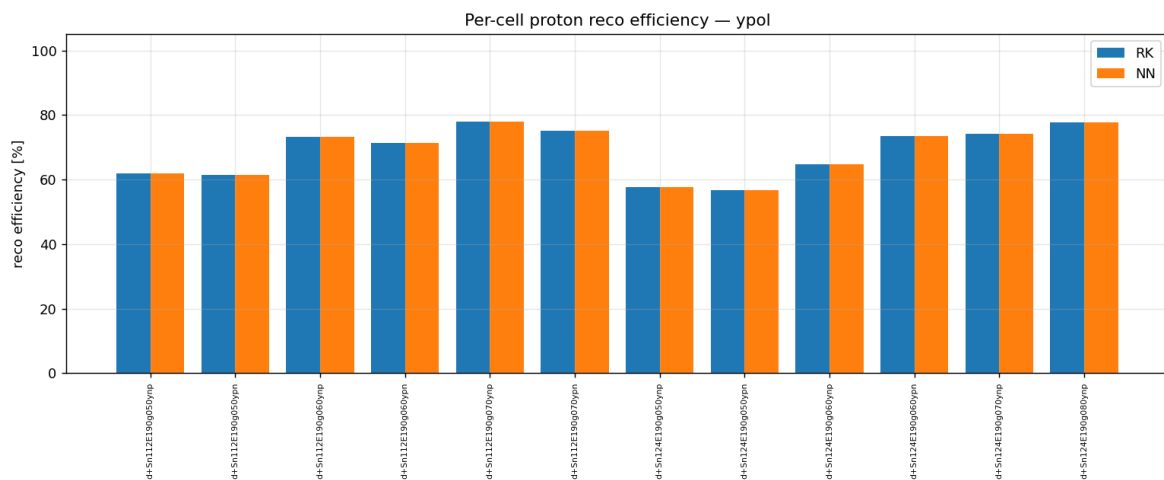


Figure 11: Per-cell proton reco efficiency, ypol.

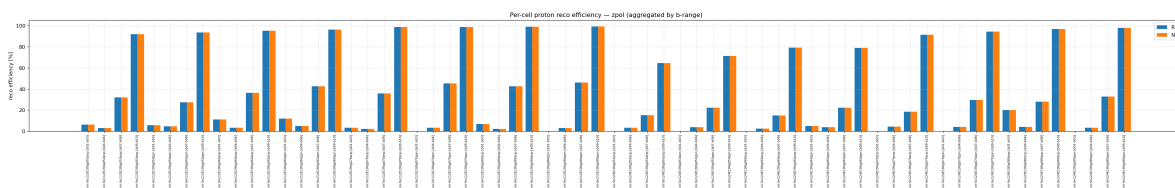


Figure 12: Per-cell proton reco efficiency, zpol (aggregated by b-range).

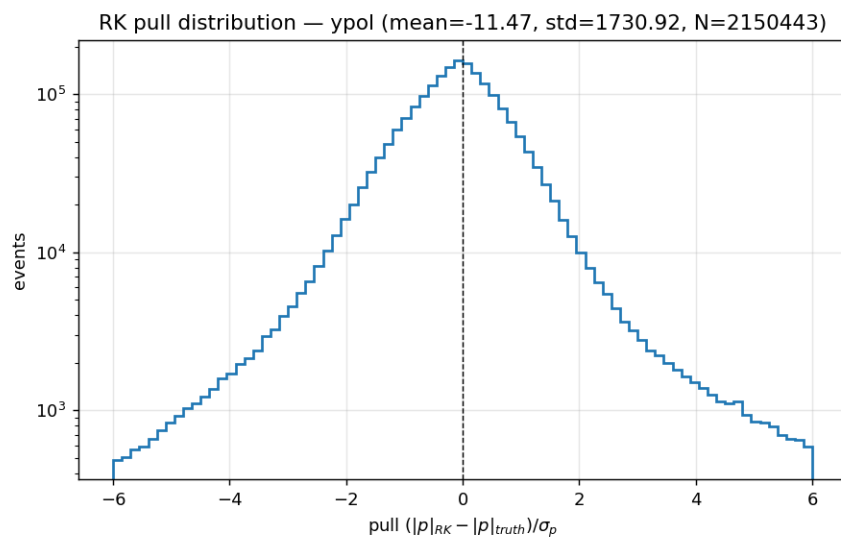


Figure 13: RK pull distribution, ypol.

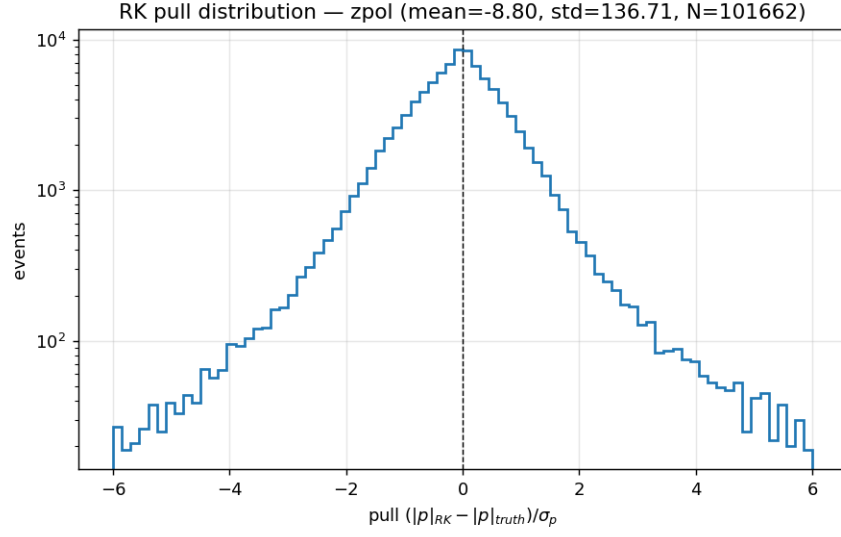


Figure 14: RK pull distribution, zpol.

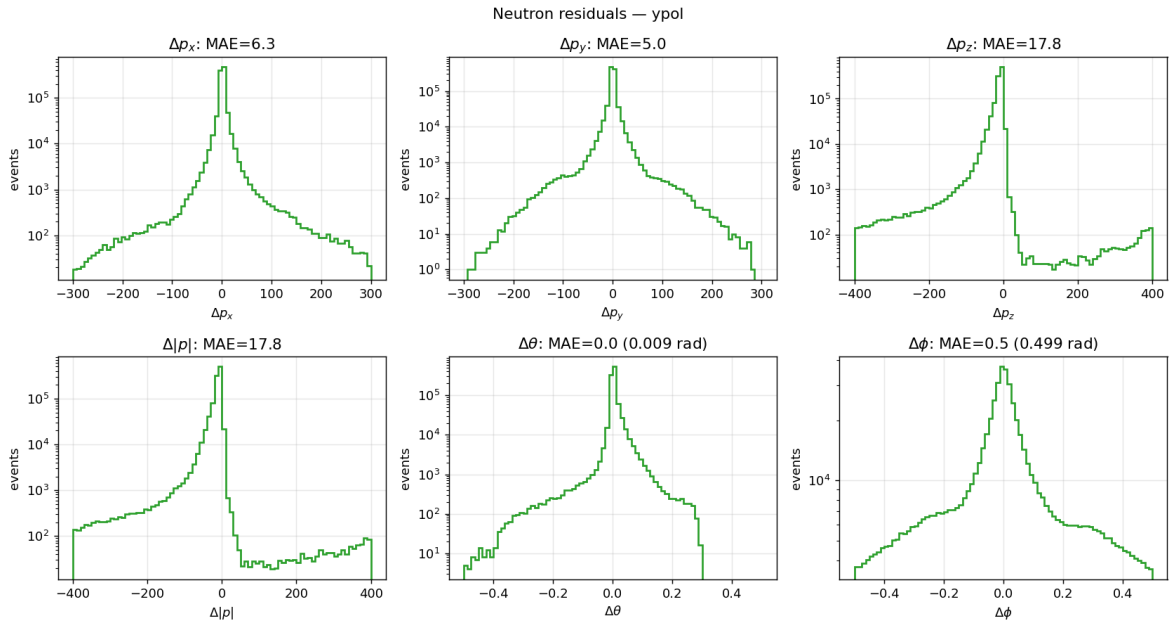


Figure 15: Neutron residual distributions, ypol.



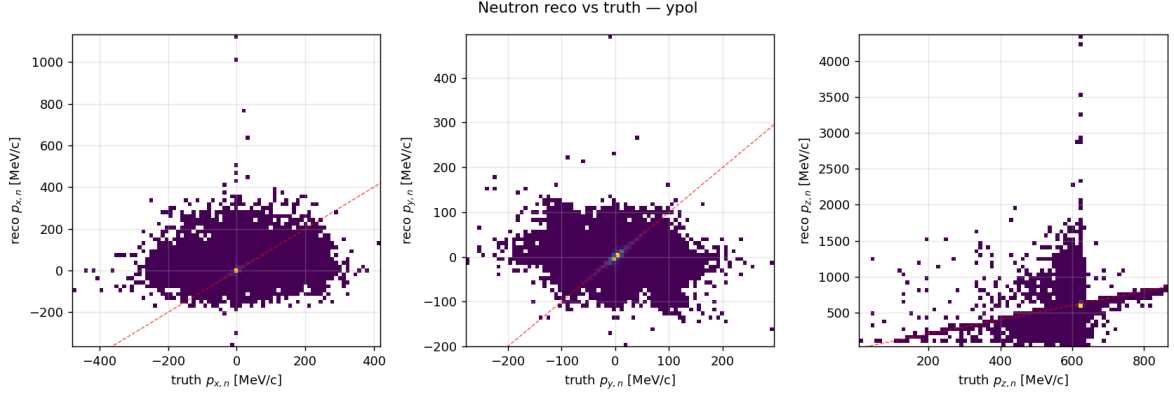


Figure 16: Neutron reco vs truth 2D scatter, ypol.

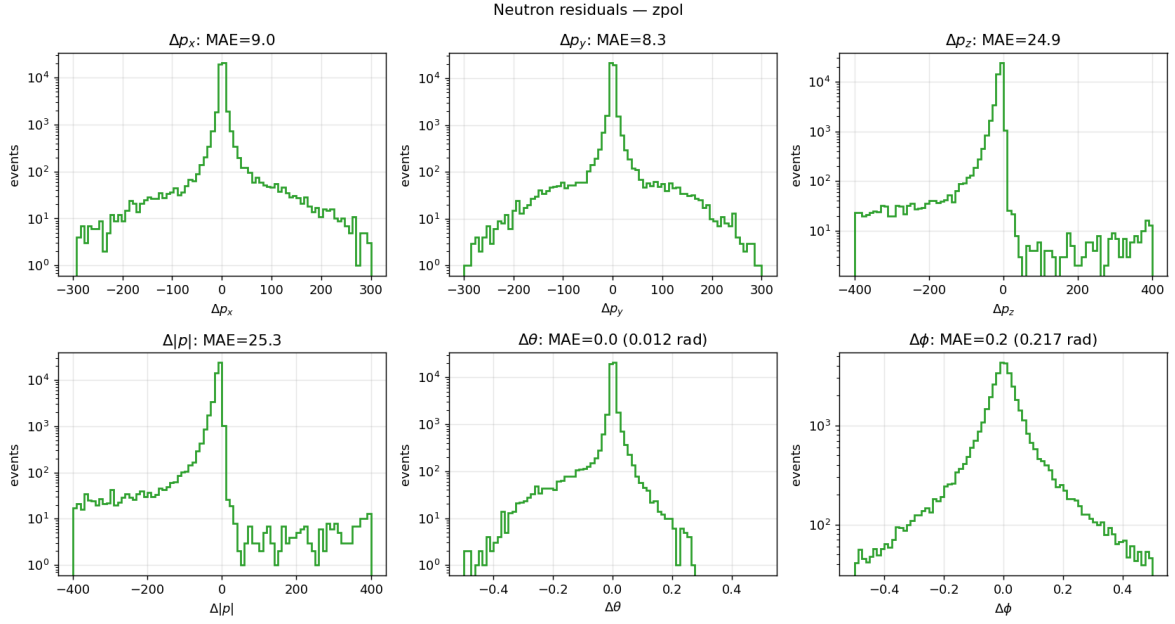


Figure 17: Neutron residual distributions, zpol.

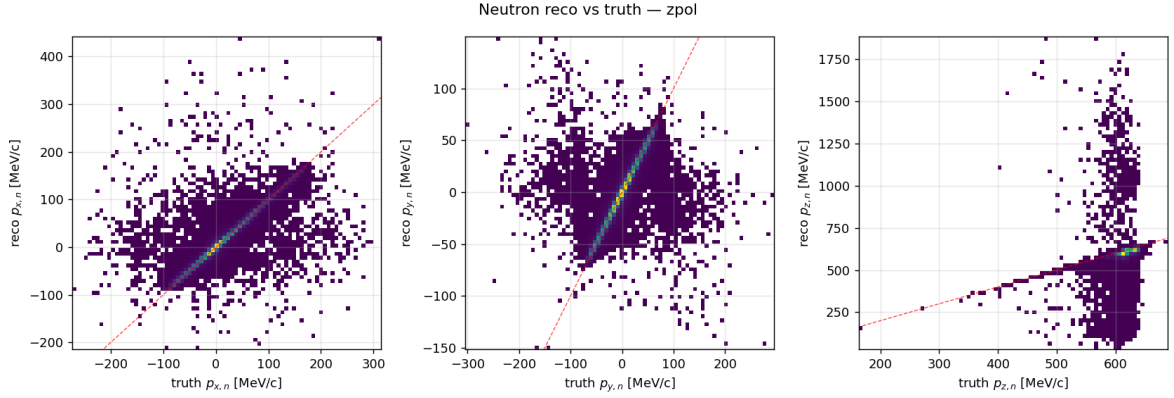


Figure 18: Neutron reco vs truth 2D scatter, zpol.